H-T-560A
September 15, 1972
SUPERSEDING
Fed. Spec. H-T-560
May 19, 1964

FEDERAL SPECIFICATION

TOOTHBRUSHES (FOR ADULTS AND CHILDREN)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers toothbrushes suitable for use by adults and children (see 6.1).

1.2 Classification. Toothbrushes covered by this specification shall be of the following types and styles as specified (see 6.2).

Type I - Adult's.
  Style A. Preventive dentistry.
  Style B. Conventional multituft.

Type II - Child's multituft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

  H-S-951 - Synthetic Filaments for Brushes; General Specifications for.
  L-P-394 - Plastic Molding Material (Propylene Plastics, Injection and Extrusion).
  PPP-B-566 - Boxes, Folding, Paperboard.
  PPP-B-636 - Box, Fiberboard.
  PPP-B-676 - Boxes, Set-Up.

Federal Standards:

  Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

  MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
  MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 1562-72 - Cellulose Propionate Molding and Extrusion Compounds.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19301.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Plastic block. The brush block (head, neck, and handle grip) shall be molded in one piece of one of the following materials at the manufacturer's option:

(a) Polypropylene in conformance with type I of L-P-304, except that the requirement and test for deformation under load shall not apply.

(b) Cellulose acetate propionate in conformance with grade H-3 of ASTM D 1562-72.

3.1.2 Synthetic filaments. Filaments shall be class N nylon, type II level, in accordance with H-S-951. The ends of the filaments shall be rounded so that sharp points and rough edges that could cause injury to the gums are eliminated. Examples of acceptably rounded filaments are shown in figure 1 (see page 3). When examined as specified in 4.3.2 not more than 25 percent of the filament ends shall fail to be acceptably rounded.

3.1.2.1 Diameter of filaments. When determined in accordance with 4.3.3, the diameter of the filaments shall be as specified in table I (see page 4) for each type and style.

3.1.3 Staple wire. Staple wire for securing the tufts in the head molding shall be nickel-silver, not less than U. S. Steel Gage No. 30 (0.012 inch in diameter) for round wire and horseshoe-style staples, and not less than 0.010 by 0.062 inch for flat wire.

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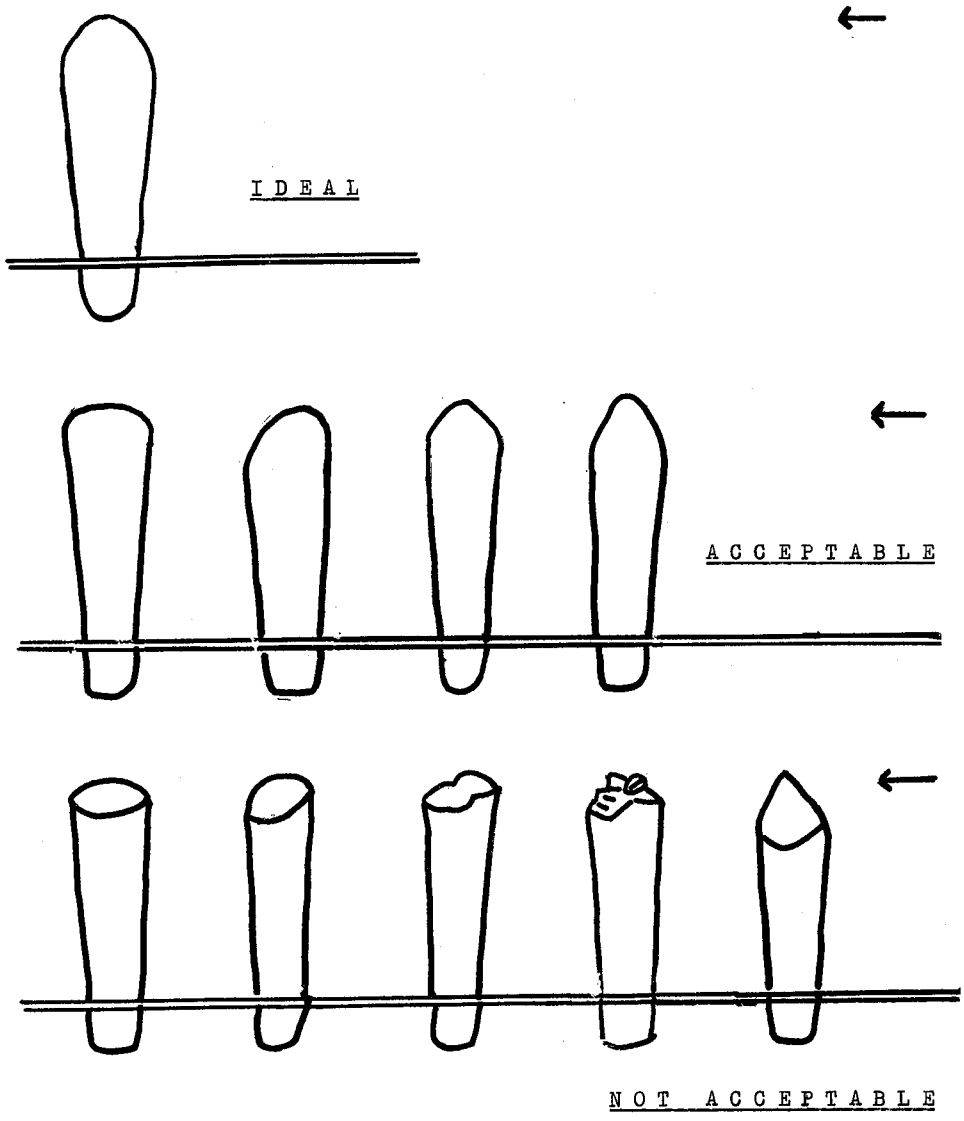


FIGURE 1. EXAMPLES OF ACCEPTABLE AND UNACCEPTABLE END-ROUNDED FILAMENTS

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3.2 Design. Toothbrushes shall be of straightline design with slightly tapered head, rounded leading end, shaped neck, and flat even working surface. The shape of the block shall be similar to figure 2.

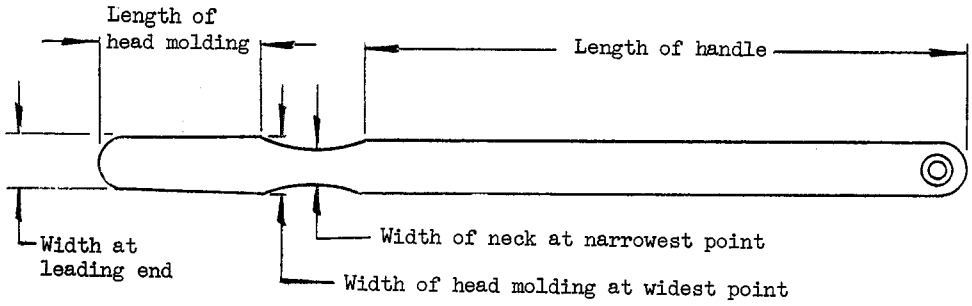


FIGURE 2. Shape of toothbrush block

3.3 Construction.

3.3.1 Blocks. The blocks of types I and II toothbrushes shall be one solid piece of molded plastic as specified in 3.1.1, of the design specified in 3.2. Differences in dimensions of type I, styles A and B, and type II toothbrushes shall be as specified in table I. Holes for tufts shall be uniformly spaced and symmetrically located in rows except that holes at the leading end may be located to follow the contour of the rounded leading end. A tapered hole 1/8 ± 1/16 inch in diameter for hanging the brush or inserting a rubber stimulator shall be located at the handle. All edges of the block shall have a radius providing a safe smooth surface.

Table I. Physical requirements for toothbrushes

Characteristics	Type I - Adult's		Type II
	Style A Preventive dentistry brush	Style B Multituft brush	Child's multituft brush
BLOCK:			
Length, overall ..... inches	6 ± 1/4	6-1/4 ± 5/16	5-3/8 ± 1/8
Head molding:			
Length (see figure 2) ..... inch...	1-1/8 ± 1/16	1-1/8 ± 1/16	1-1/16 ± 1/32
Width at widest place (see figure 2).. do....	7/16 ± 1/32	7/16 ± 1/32	23/64 ± 1/64
Width at leading end (see figure 2)... do....	13/32 ± 1/32	13/32 ± 1/32	21/64 ± 1/64
Thickness, minimum	3/16	3/16	3/16
Number of holes for tufts	18	42-43	27-30
Number of rows of holes	3	4	3
Number of holes in each row	6	(see note a)	(see note b)
Neck:			
Width at narrowest place, minimum ..... inch..	7/32	7/32	7/32
Thickness, minimum ..... do.	3/16	3/16	3/16
Handle:			
Length (see figure 2) ..... inches	3-3/4 ± 1/4	3-3/4 ± 1/4	3-5/8 ± 1/8
Width, inch ..... inch	1/2 ± 1/16	1/2 ± 1/16	7/16 ± 1/16
Thickness, minimum ..... do.	5/32	5/32	5/32
FILAMENTS:			
Number of filament ends per tuft			
Minimum	78	30	30
Maximum	84	34	34
Average	82	32	32
Length clear of block .....inches	7/16 ± 1/32	7/16 ± 1/32	3/8 ± 1/32
Diameter .....inch	0.007 ± 0.0006	0.008 ± 0.0007	0.008 ± 0.0007

a. Type I, style B toothbrushes shall have four rows of ten holes each, plus two or three at the end.
b. Type II children's toothbrushes shall have three rows of 8 or 9 holes each. Three additional holes may be added at the end.

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3.3.2 Brush part. The tufts of filaments shall solidly fill the holes in the face of the head molding. A staple or slug (see 3.1.3) spanning the filaments in each tuft midway between the ends and driven into the bottom of each hole shall securely anchor the filaments in the head molding. There shall be no cracks in the block due to the stapling operation. The rounded filament ends (see 3.1.2) shall form a flat even working surface. Filaments shall be splayed not more than 15 degrees from a line perpendicular to the axis of the block.

3.4 Filament count per tuft. When determined as specified in 4.3.1, the number of filament ends per tuft shall be in conformance with the applicable requirement specified in table I.

3.5 Resistance of brushes to boiling water. After immersion in boiling water, 100° C. (212° F.) at sea level, for not less than 20 minutes and drying at room temperature, 27° ± 5° C. (80° ± 9° F.), for at least 24 hours as specified in 4.3.4, brush blocks (head, neck, and handle grip) shall not discolor or become distorted and filaments shall not change in color or stiffness.

3.6 Identification of nylon filaments. In case of dispute, nylon filaments in the finished brush shall meet the following additional requirements.

3.6.1 Specific gravity. When tested in monochlorobenzene (specific gravity 1.10) as specified in 4.3.5.1, nylon filaments from the finished brush shall float in the liquid medium.

3.6.2 Resistance to concentrated hydrochloric acid. When tested in concentrated hydrochloric acid as specified in 4.3.5.2, nylon filaments from the finished brush shall not completely dissolve.

3.7 Color.

3.7.1 Blocks. Unless otherwise specified, the color of toothbrush blocks shall be the manufacturer's standard colors (see 6.2).

3.8 Workmanship. Brushes shall be clean and free from defects that affect serviceability, durability, or appearance. Occurrence of defects shall not exceed the quality level specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of quality. Certificates of quality supplied by the manufacturer of the synthetic filament and plastic block may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the manufacturer, the name of the material, the lot number, lot size, sample size, date of testing, test method, individual test results, and the specification requirements. Where certificates of compliance are submitted the Government reserves the right to check test such items to determine the validity of the certificates.

4.2 Sampling for inspection (examination and testing) and acceptance. Sampling for examination, testing, and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be examined and tested in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item examination and testing.

4.2.2.1 Lot. For the purpose of examination and testing of finished brushes, a lot shall, so far as is practicable, consist of all brushes made of the same type and style manufactured under essentially the same conditions at essentially the same time.

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4.2.2.2 Examination for defects other than dimensional. Examination shall be made of the end item for defects listed in table II. The inspection level shall be II and the acceptable quality level (AQL) 4.0 defects per 100 units. The sample unit shall be one brush. If the number of defects in the sample for examination exceeds the acceptance number specified in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of the specification.

TABLE II. Defects other than dimensional

Examination	Defects
Construction:	
Block (handle, neck and head)	Not one-piece solid plastic. Surfaces not smooth; edges and corners sharp - not rounded; rough, pitted, chipped, nicked, or cracked surfaces. Color not as required (when specified). Deformed or distorted. Dirty - embedded with foreign material. Hole in handle missing. Unpleasant odor detected.
Filaments	Not natural color (undyed) nylon. Soiled or embedded with foreign material. Loose - can be easily removed by hand. Less than required number of rows and tufts in each row. Filaments form an uneven working surface.
Set of tufts	Not firmly anchored with slugs or staples - can be easily removed by hand.
Workmanship	Defects present which affect appearance, serviceability, or durability.

4.2.2.3 Dimensional examination. Examination shall be made of the end item for compliance with the dimensional requirements in table I as applicable. Inspection level shall be S-2 and acceptable quality level (AQL) 4.0 percent defective. The sample unit shall be one brush. If the sample unit fails to meet any dimensional requirement specified in table I, it shall be considered defective. If the number of defective sample units in the sample exceeds the acceptance number specified in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of the specification.

4.2.2.4 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of sample units offered for inspection at one time. The inspection level shall be S-2 with and the acceptable quality level (AQL) 4.0 expressed in terms of defects per hundred units.

Table III. Examination of preparation for delivery

Examination	
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application. Component missing or damaged.
Materials	Bulging or distortion of containers. Cushioning inadequate, improper, or missing.

4.2.2.4.1 Examination of closure sealing and reinforcement of containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure sealing and reinforcement shall be in accordance with the appendix of that specification.

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4.2.2.5 Testing. The end item shall be tested as specified in 4.3 through 4.3.5.2. The inspection level shall be S-1 and the acceptable quality level (AQL) 4.0 percent defective. The sample unit shall be two brushes. If any brush in the sample unit fails to meet any requirement for which tests are made, the sample unit shall be considered defective. If the number of defective sample units in the sample exceeds the acceptance number in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of the specification.

4.3 Tests for end item. Unless otherwise specified herein, tests on the end item shall be conducted at room temperature, 27° ± 5° C. (80° ± 9° F.), in accordance with the following procedures:

4.3.1 Tuft count. Carefully remove six tufts from representative places in the brush part. Count the number of filaments in each tuft and record each value separately. Check each result for conformance with the applicable requirement in table I. Calculate the average value for the six tufts and check the results for conformance with the applicable requirements in table I. Calculate the total number of filaments in the six tufts and record the result as A.

4.3.2 Examination of filaments for end rounding. Examination of the filament ends shall be made using a binocular microscope having a 10-power (10X) lens and a setting of 1.5 (1.5X magnification), or other suitable apparatus with equivalent minimum magnification. When using a binocular microscope, the filament ends in not less than 6 tufts selected from representative locations in the brush head shall be examined for end rounding, the percent defective filaments determined by dividing defective by total filaments, and the results recorded and checked for conformance with 3.1.2. When an instrument such as a comparator is used which permits examination of the entire brushing surface, the percentage of defective filaments shall be noted and checked for conformance with 3.1.2.

4.3.3 Diameter of filaments. The average diameter of the filaments in the finished brush shall be determined by the applicable method in H-S-951 and the result checked for conformance with the applicable requirement in table I. The specimen shall be composed of filaments from tufts located at representative places in the brush part.

4.3.4 Resistance of brush to boiling water. One brush from each test unit shall be tested as follows: Place the brush in a container of water adjusted to the required temperature and allow it to remain in the water for the required period of time (see 3.5), taking care that the brush is completely submerged and that water circulates freely around it during the immersion period. At the end of the immersion period, remove the brush from the water and allow it to dry for the required period of time (see 3.5) and examine it for compliance with 3.5.

4.3.5 Identification of nylon filaments in the end item. In case of dispute regarding the identity of the filaments in the finished toothbrush, filaments shall be tested as specified in 4.3.5.1 and 4.3.5.2. Remove 5 tufts from representative places in the brush part and reserve the filaments in each tuft for the specified tests.

4.3.5.1 Specific gravity. Take two filaments from each of the five removed tufts in 4.3.5. Place the 10 filaments in a suitable glass container holding sufficient monochlorobenzene (specific gravity 1.10) to permit the filaments to circulate freely in the solution. With a suitable instrument poke the filaments beneath the surface of the solvent to thoroughly wet them. Observe each filaments for conformance with 3.6.1.

4.3.5.2 Resistance to hydrochloric acid. Place 10 filaments, two from each of the five removed tufts in 4.3.5, in a suitable glass container. Cover the filaments with concentrated hydrochloric acid (37-38 percent, specific gravity 1.19) and allow them to remain immersed in the acid for one hour. Care should be taken to prevent the acid from coming into contact with the skin or clothing. Examine the filaments for conformance with 3.6.2.

5 PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level B or C, as specified (see 6.2).

5.1.1 Level B, unit packaging. Each toothbrush shall be packaged to preserve cleanliness specified in 3.10, by one of the following methods: (a) a wrapper of cellophane or plastic film, (b) a plastic container, or (c) a container overwrapped with cellophane or plastic film.

5.1.1.1 Intermediate packaging. Twelve toothbrushes of one type and style only, packaged as specified in 5.1, shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Each box shall be secured to prevent accidental opening.

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5.1.2 Level C. Toothbrushes shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Toothbrushes of one type and style only, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, grade V3c or V3s. Each shipping container shall be closed sealed and reinforced in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Toothbrushes of one type and style only, packaged as specified in 5.1, shall be packed in quantities as specified (see 6.2), in a box conforming to PPP-B-636, class domestic.

5.2.3 Level C. Toothbrushes, packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special markings required in procurement documents, commercial markings are acceptable on the unit product or unit package. Marking of the unit product or unit package, intermediate, and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use.

6.1.1 Type I, style A - preventive dentistry brush. Preventive dentistry toothbrushes are extra soft brushes for use in preventing and arresting conditions which create dental problems, by removing food particles and bacterial plaque from teeth and gums without injury to either.

6.1.2 Type I, style B - conventional multituft brush. Conventional multituft toothbrushes are intended for use by adults for general everyday brushing of teeth.

6.1.3 Type II, child's multituft brush. The child's toothbrush described in this specification is intended for use by adults in brushing the teeth of very small children, or by children old enough to properly perform the brushing operation.

6.2 Ordering data. Purchasers should exercise any options offered herein (see 3.7.1) and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style required (see 1.2.1).
(c) Applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Quantity to shipping container (see 5.2.2)
(e) Special marking required (see 5.3).

CIVILIAN AGENCIES:

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